

NORTHEASTERN, NC, USA

WMO: 723074

Lat: 36.028N

Lon: 76.567W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03703

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-2.8	-16.1	0.9	-0.3	-13.0	1.2	0.6	10.4	9.1	9.0	8.1	2.2	0	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	34.1	25.5	32.9	25.3	32.1	24.9	27.3	31.8	26.5	30.7	25.9	29.8	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.6	29.6	25.3	20.5	28.7	24.8	19.9	28.3	85.7	31.6	82.5	30.6	80.2	29.9	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.6	-8.5	36.0	1.9	1.5	-9.8	37.1	-10.9	38.0	-12.0	38.9	-13.4	40.0		
(5)			-9.5	28.2	1.8	1.3	-10.8	29.2	-11.9	29.9	-12.9	30.6	-14.3	31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	6.2	7.5	10.8	16.2	20.9	25.1	26.9	26.3	23.0	17.6	11.8	8.1		
	DBStd	8.55	5.71	5.16	5.20	4.70	4.07	3.13	2.59	2.50	3.19	4.53	4.67	4.87		
	HDD10.0	439	148	102	54	4	0	0	0	0	0	2	32	96		
	HDD18.3	1635	378	306	237	96	33	1	0	0	4	70	202	318		
	CDD10.0	2903	29	31	81	190	237	454	525	507	389	237	86	37		
	CDD18.3	1057	1	2	6	31	102	205	267	248	143	46	5	1		
	CDH23.3	9384	4	9	37	219	749	1871	2769	2403	1032	270	19	2		
	CDH26.7	3194	0	0	3	44	211	663	1084	868	269	49	1	0		
(14) Wind	WSAvg	2.9	3.0	3.2	3.4	3.6	3.1	2.9	2.7	2.4	2.5	2.5	2.5	2.6		
(15) Precipitation	PrecAvg	1229	100	86	104	82	105	115	146	135	116	89	69	74		
	PrecMax	1709	233	200	220	178	250	278	348	300	479	277	146	172		
	PrecMin	830	9	27	25	16	23	24	48	23	5	0	21	9		
	PrecStd	195	47	43	45	39	55	53	73	70	91	70	33	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	24.9	26.6	30.2	32.9	35.0	36.1	35.2	32.6	30.1	25.9	22.8		
		MCWB	19.3	18.5	17.5	21.3	24.1	25.1	26.5	26.3	24.1	23.7	19.5	18.2		
	2%	DB	20.0	21.7	24.0	27.7	30.9	33.1	34.3	33.7	31.1	27.9	22.9	20.9		
		MCWB	16.2	16.0	16.6	19.4	22.9	24.4	26.2	25.8	24.0	21.8	18.1	17.8		
	5%	DB	17.5	18.7	22.1	26.1	28.9	32.2	32.9	32.4	29.4	26.3	21.6	18.1		
		MCWB	14.6	14.7	16.0	18.9	22.1	24.3	25.7	25.6	23.8	21.0	17.7	16.1		
	10%	DB	15.6	16.3	19.4	23.9	27.5	30.6	32.0	31.1	27.9	24.6	19.4	16.5		
		MCWB	13.0	12.6	15.2	18.1	21.2	23.7	25.4	24.8	23.1	20.5	16.5	13.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	19.6	20.0	23.1	25.2	26.9	28.7	28.0	26.6	25.2	21.4	19.7		
		MCDB	21.9	23.2	23.7	27.1	30.0	31.3	34.3	33.2	29.8	27.5	23.7	21.5		
	2%	WB	17.4	17.5	18.6	21.6	24.1	26.0	27.5	27.1	25.8	24.0	19.9	18.2		
		MCDB	19.0	19.8	22.1	25.2	28.5	30.7	32.1	31.7	28.7	26.4	22.2	19.6		
	5%	WB	15.3	15.5	17.2	20.4	23.3	25.3	26.8	26.4	25.1	22.8	18.4	16.6		
		MCDB	17.2	18.2	20.1	24.2	27.4	29.8	31.0	30.4	28.0	25.1	20.6	18.2		
	10%	WB	13.2	13.1	15.8	19.1	22.5	24.7	26.1	25.7	24.4	21.3	16.9	14.5		
		MCDB	15.0	15.8	19.0	22.9	26.3	28.9	30.1	29.5	27.2	23.7	18.7	16.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.4	10.8	10.9	9.7	8.9	8.3	8.2	8.6	10.1	10.6	9.8		
		MCDBR	11.9	13.3	13.0	12.2	10.6	9.7	9.0	9.0	9.1	10.4	11.3	11.7		
	5% WB	MCWBR	9.3	8.9	7.2	5.9	4.5	3.9	3.9	3.8	4.0	5.3	7.4	9.4		
		MCWBR	11.1	12.4	10.9	10.5	9.5	8.6	8.2	8.1	7.2	8.0	9.7	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.324	0.350	0.391	0.438	0.493	0.521	0.509	0.413	0.360	0.327	0.315		
	taud		2.539	2.490	2.444	2.337	2.238	2.131	2.067	2.091	2.360	2.494	2.535	2.552		
	Ebn at Noon		895	918	919	893	851	801	776	776	842	865	864	863		
	Edh at Noon		79	93	106	124	139	155	164	158	114	92	79	74		
(44) All-Sky Solar Radiation	RadAvg		2.49	3.23	4.24	5.41	6.00	6.33	6.00	5.43	4.47	3.66	2.78	2.22		
	RadStd		0.18	0.35	0.31	0.39	0.46	0.34	0.31	0.32	0.32	0.43	0.21	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+163	+110

Nomenclature: See separate page